Cloud Lab - 8

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Roll no: 2023BCD0059

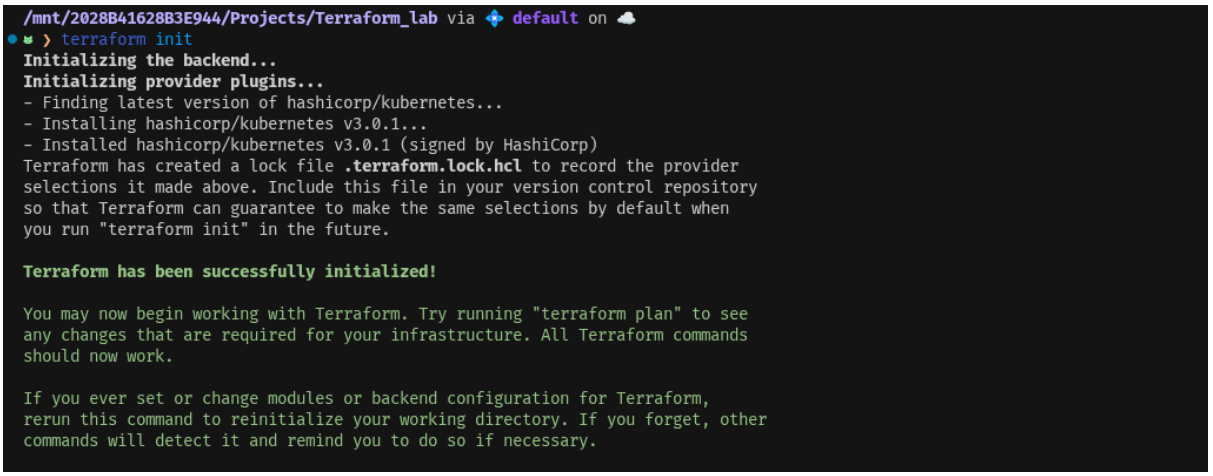
1)

main.tf

```
provider "kubernetes" {
  config_path = "~/.kube/config"
}

resource "kubernetes_namespace" "lab" {
  metadata {
    name = "lab-namespace"
  }
}

resource "kubernetes_deployment" "nginx" {
  metadata {
    name      = "nginx-deployment"
    namespace = kubernetes_namespace_v1.lab.metadata[0].name
    labels = {
      app = "nginx"
    }
  }

  spec {
    replicas = 2

    selector {
      match_labels = {
        app = "nginx"
      }
    }

    template {
      metadata {
        labels = {
          app = "nginx"
        }
      }

      spec {
        container {
          image = "nginx:latest"
          name  = "nginx"
```

```
        port {
          container_port = 80
        }
      }
    }
  }
 }
}
```

Creates two pods that run nginx on port 80. If one dies it auto recreates the other one.



```
❯ terraform apply
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # kubernetes_deployment_v1.nginx will be created
  + resource "kubernetes_deployment_v1" "nginx" {
      + id               = (known after apply)
      + wait_for_rollout = true

      + metadata {
          + generation       = (known after apply)
          + labels           = {
              + "app" = "nginx"
            }
          + name             = "nginx-deployment"
          + namespace        = "lab-namespace"
          + resource_version = (known after apply)
          + uid              = (known after apply)
        }

      + spec {
          + min_ready_seconds         = 0
          + paused                    = false
          + progress_deadline_seconds = 600
          + replicas                  = "2"
          + revision_history_limit    = 10

          + selector {
              + match_labels = {
                  + "app" = "nginx"
                }
            }

          + strategy (known after apply)

          + template {
              + metadata {
                  + generation       = (known after apply)
                  + labels           = {
                      + "app" = "nginx"
                    }
                  + name             = (known after apply)
                  + resource_version = (known after apply)
                  + uid              = (known after apply)
                }
              + spec {
                  + automount_service_account_token  = true
                  + dns_policy                       = "ClusterFirst"
                  + enable_service_links             = true
                  + host_ipc                         = false
```

```
❯ terraform apply
                  + host_ipc                      = false
                  + host_network                  = false
                  + host_pid                      = false
                  + hostname                      = (known after apply)
                  + node_name                     = (known after apply)
                  + restart_policy                = "Always"
                  + scheduler_name                = (known after apply)
                  + service_account_name          = (known after apply)
                  + share_process_namespace       = false
                  + termination_grace_period_seconds = 30

                  + container {
                      + image                      = "nginx:latest"
                      + image_pull_policy          = (known after apply)
                      + name                       = "nginx"
                      + restart_policy             = (known after apply)
                      + stdin                      = false
                      + stdin_once                 = false
                      + termination_message_path   = "/dev/termination-log"
                      + termination_message_policy = (known after apply)
                      + tty                        = false

                      + port {
                          + container_port = 80
                          + protocol       = "TCP"
                        }

                      + resources (known after apply)
                    }

                  + image_pull_secrets (known after apply)

                  + readiness_gate (known after apply)
                }
            }
        }
    }

  # kubernetes_namespace_v1.lab will be created
  + resource "kubernetes_namespace_v1" "lab" {
      + id                                 = (known after apply)
      + wait_for_default_service_account = false

      + metadata {
          + generation       = (known after apply)
          + name             = "lab-namespace"
          + resource_version = (known after apply)
          + uid              = (known after apply)
        }
    }
```

```
❯ terraform apply
                    }

                  + image_pull_secrets (known after apply)

                  + readiness_gate (known after apply)
                }
            }
        }
    }

  # kubernetes_namespace_v1.lab will be created
  + resource "kubernetes_namespace_v1" "lab" {
      + id                                 = (known after apply)
      + wait_for_default_service_account = false

      + metadata {
          + generation       = (known after apply)
          + name             = "lab-namespace"
          + resource_version = (known after apply)
          + uid              = (known after apply)
        }
    }

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

kubernetes_namespace_v1.lab: Creating...
kubernetes_namespace_v1.lab: Creation complete after 0s [id=lab-namespace]
kubernetes_deployment_v1.nginx: Creating...
kubernetes_deployment_v1.nginx: Still creating... [00m10s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [00m20s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [00m30s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [00m40s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [00m50s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [01m00s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [01m10s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [01m20s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [01m30s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [01m40s elapsed]
kubernetes_deployment_v1.nginx: Still creating... [01m50s elapsed]
kubernetes_deployment_v1.nginx: Creation complete after 1m56s [id=lab-namespace/nginx-deployment]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

/mnt/2028B41628B3E944/Projects/Terraform_lab via 💠 default on ☁️  took 2m31s
❯
```

```
/mnt/2028B41628B3E944/Projects/Terraform_lab via 💠 default on ☁️  took 2m31s
❯ kubectl get pods -n lab-namespace
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-78df899b6f-6vwk4   1/1     Running   0          6m31s
nginx-deployment-78df899b6f-zgzh8   1/1     Running   0          6m31s

/mnt/2028B41628B3E944/Projects/Terraform_lab via 💠 default on ☁️
❯
```

2)

Creating a mongo docker container

```
fish /home/harisshragav

~ via 🐍 v3.13.5 on ☁️
❯ docker run -d \
        --name mongo-container \
        -p 27017:27017 \
        mongo:latest
Unable to find image 'mongo:latest' locally
latest: Pulling from library/mongo
01a854eada6f: Pull complete
e8567d1c4785: Pull complete
1e41d5f93e35: Pull complete
3a5c9bc3c6a6: Pull complete
817807f3c64e: Pull complete
2605962b0286: Pull complete
0c85015575ad: Pull complete
8d1d6859a473: Pull complete
ee7db4c3583f: Download complete
51deed191cab: Download complete
Digest: sha256:d343c378b5c6e2fe373174abcf4a877be0dfc721b5d0b9d582204dccb1c00b86
Status: Downloaded newer image for mongo:latest
ad4888773070e23b9593bfd4554733cb8b92e02109f916ef2233d56446660af0
```

2) Creating a mongo docker container

Creating a node app to perform crud operation on the mogo server

[app.js](app.js)

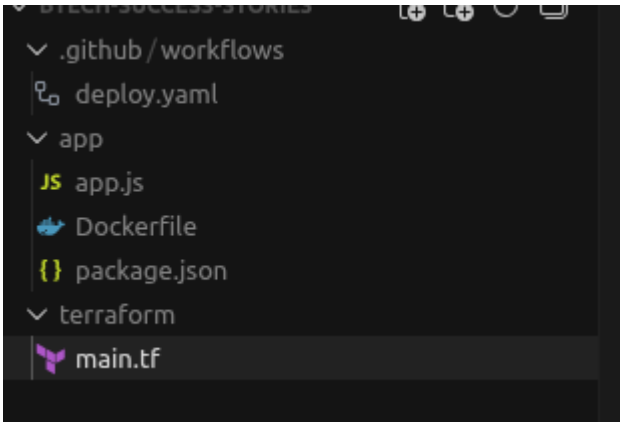
```
const { MongoClient } = require("mongodb");

const uri = "mongodb://localhost:27017";
const client = new MongoClient(uri);

async function run() {
 await client.connect();

 const db = client.db("btechStories");
 const collection = db.collection("success");

 // CREATE (multiple)
 await collection.insertMany([
   {
     name: "Harissh",
     year: "3rd Year",
     achievement: "Built multi-agent system",
     mood: "🔥"
   },
   {
     name: "Aarav",
     year: "2nd Year",
     achievement: "Won hackathon",
```

```
    mood: "🚀"
  },
  {
    name: "Diya",
    year: "4th Year",
    achievement: "Published research paper",
    mood: "📄"
  },
  {
    name: "Rohan",
    year: "1st Year",
    achievement: "Started coding journey",
    mood: "💻"
  }
]);

console.log("Multiple records inserted!");

// READ
const data = await collection.find().toArray();
console.log("All Records:", data);

// UPDATE (update one person)
await collection.updateOne(
  { name: "Aarav" },
  { $set: { mood: "🏆 winner energy" } }
);

console.log("Updated Aarav!");

// DELETE (delete one person)
await collection.deleteOne({ name: "Rohan" });

console.log("Deleted Rohan!");

await client.close();
}

run();
```

```
/mnt/2028B41628B3E944/Projects/mongo-node-app is 📦 v1.0.0 via ⬢ v24.6.0 on ☁️  took 14s
❯ node app.js
Multiple records inserted!
All Records: [
  {
    _id: new ObjectId('69ba48d8676087e8ce3bc257'),
    name: 'Harissh',
    year: '3rd Year',
    achievement: 'Built multi-agent system',
    mood: '🔥'
  },
  {
    _id: new ObjectId('69ba48d8676087e8ce3bc258'),
    name: 'Aarav',
    year: '2nd Year',
    achievement: 'Won hackathon',
    mood: '🚀'
  },
  {
    _id: new ObjectId('69ba48d8676087e8ce3bc259'),
    name: 'Diya',
    year: '4th Year',
    achievement: 'Published research paper',
    mood: '📄'
  },
  {
    _id: new ObjectId('69ba48d8676087e8ce3bc25a'),
    name: 'Rohan',
    year: '1st Year',
    achievement: 'Started coding journey',
    mood: '💻'
  }
]
Updated Aarav!
Deleted Rohan!
```

3)

File Structure:



app.js

```
const express = require("express");
const { MongoClient } = require("mongodb");

const app = express();
app.use(express.json());
```

```
const uri = "mongodb://mongodb-service:27017";
const client = new MongoClient(uri);

let collection;

async function init() {
 await client.connect();
 const db = client.db("btechStories");
 collection = db.collection("success");
}

app.post("/create", async (req, res) => {
 const result = await collection.insertOne(req.body);
 res.send(result);
});

app.get("/read", async (req, res) => {
 const data = await collection.find().toArray();
 res.send(data);
});

app.put("/update", async (req, res) => {
 await collection.updateOne(
   { name: req.body.name },
   { $set: { mood: req.body.mood } }
 );
 res.send("Updated");
});

app.delete("/delete", async (req, res) => {
 await collection.deleteOne({ name: req.body.name });
 res.send("Deleted");
});

init().then(() => {
 app.listen(3000, () => console.log("Server running"));
});
```

Dockerfile

```
FROM node:18

WORKDIR /app
```

```
COPY package.json .
RUN npm install

COPY . .

CMD ["node", "app.js"]
```

main.tf

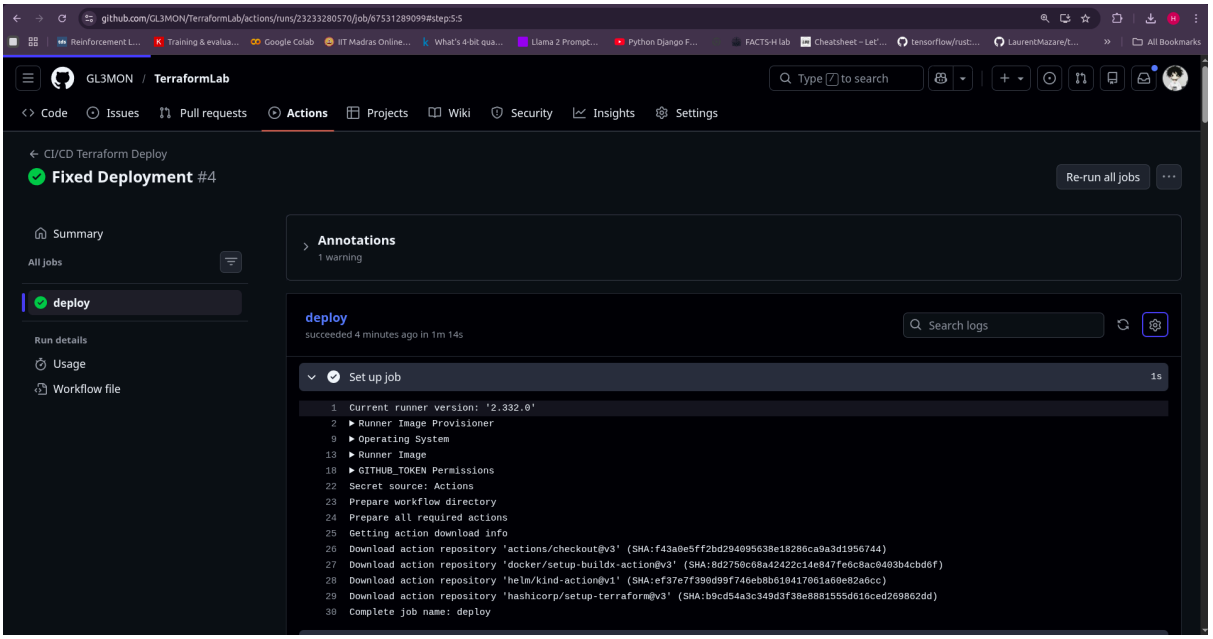
```
provider "kubernetes" {
  config_path = "~/.kube/config"
}

resource "kubernetes_namespace_v1" "btech-success-stories" {
  metadata {
    name = "btech-success-stories"
  }
}

# --------------------
# MongoDB Deployment
# --------------------
resource "kubernetes_deployment" "mongodb" {
  metadata {
    name      = "mongodb"
    namespace = kubernetes_namespace_v1.btech-success-stories.metadata[0].name
  }

  spec {
    replicas = 1

    selector {
      match_labels = {
        app = "mongodb"
      }
    }

    template {
      metadata {
        labels = {
          app = "mongodb"
        }
      }

      spec {
        container {
          name  = "mongodb"
          image = "mongo:latest"
```

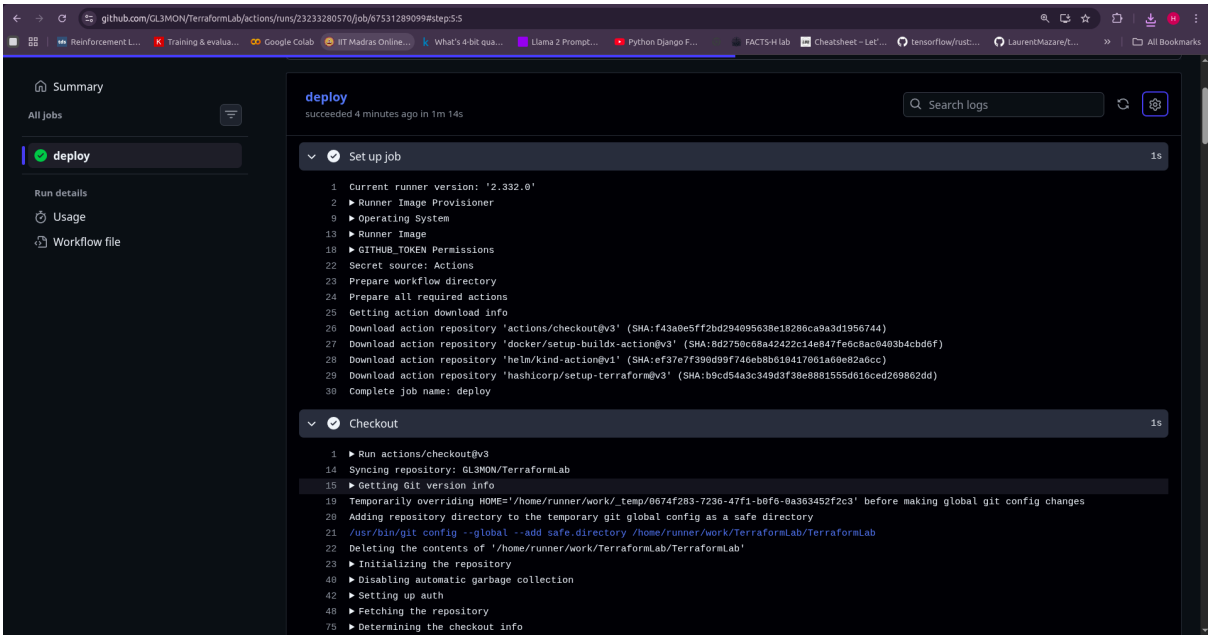
```
        port {
          container_port = 27017
        }
      }
    }
  }
 }
}

# Mongo Service
resource "kubernetes_service_v1" "mongodb_service" {
  metadata {
    name      = "mongodb-service"
    namespace = kubernetes_namespace_v1.btech-success-stories.metadata[0].name
  }

  spec {
    selector = {
      app = "mongodb"
    }

    port {
      port        = 27017
      target_port = 27017
    }
  }
}
```

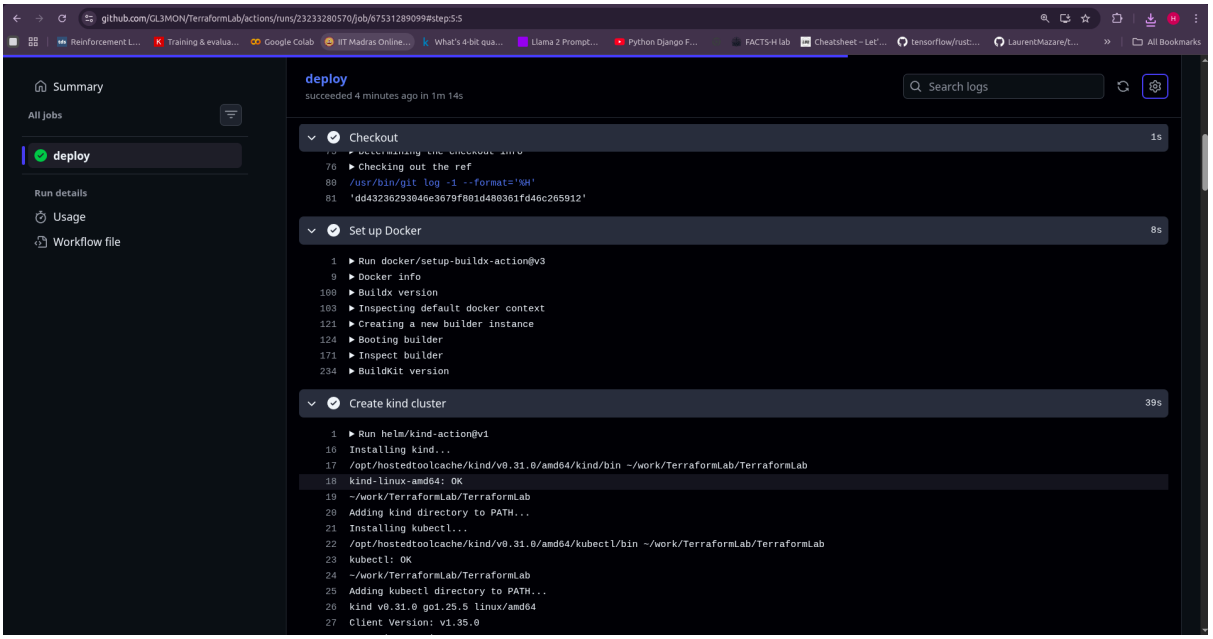
deployment.yaml

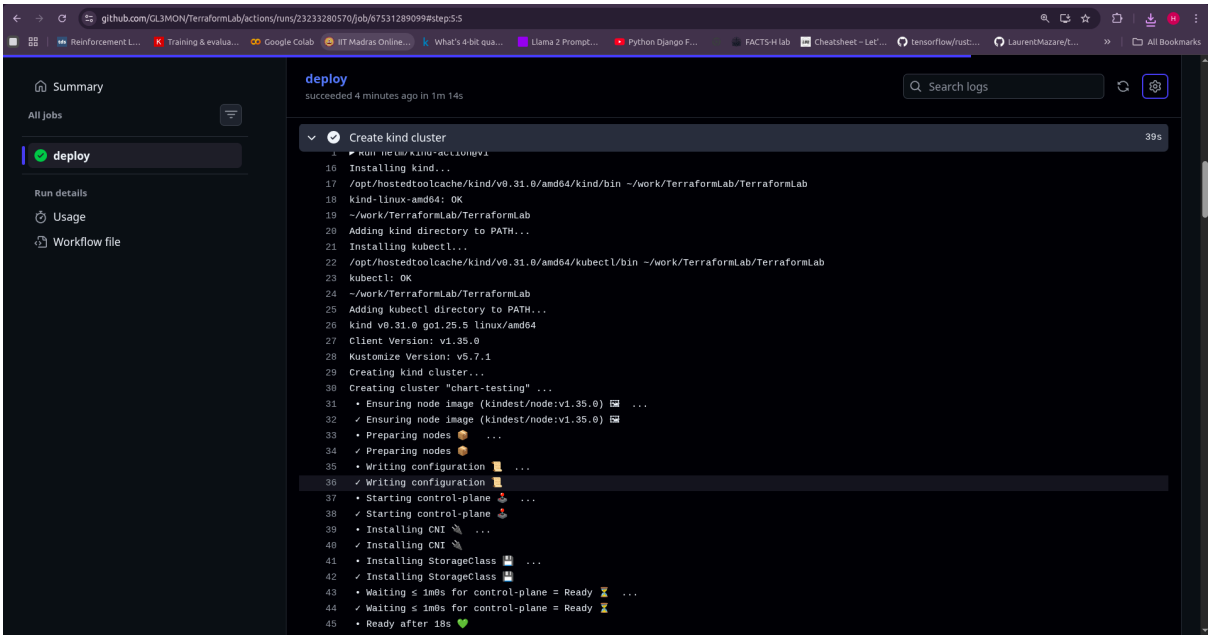
```yaml
name: CI/CD Terraform Deploy

on:
 push:
   branches:
     - main

jobs:
 deploy:
   runs-on: ubuntu-latest

   steps:
     - name: Checkout
       uses: actions/checkout@v3

     # Install Docker (already available, but safe)
```

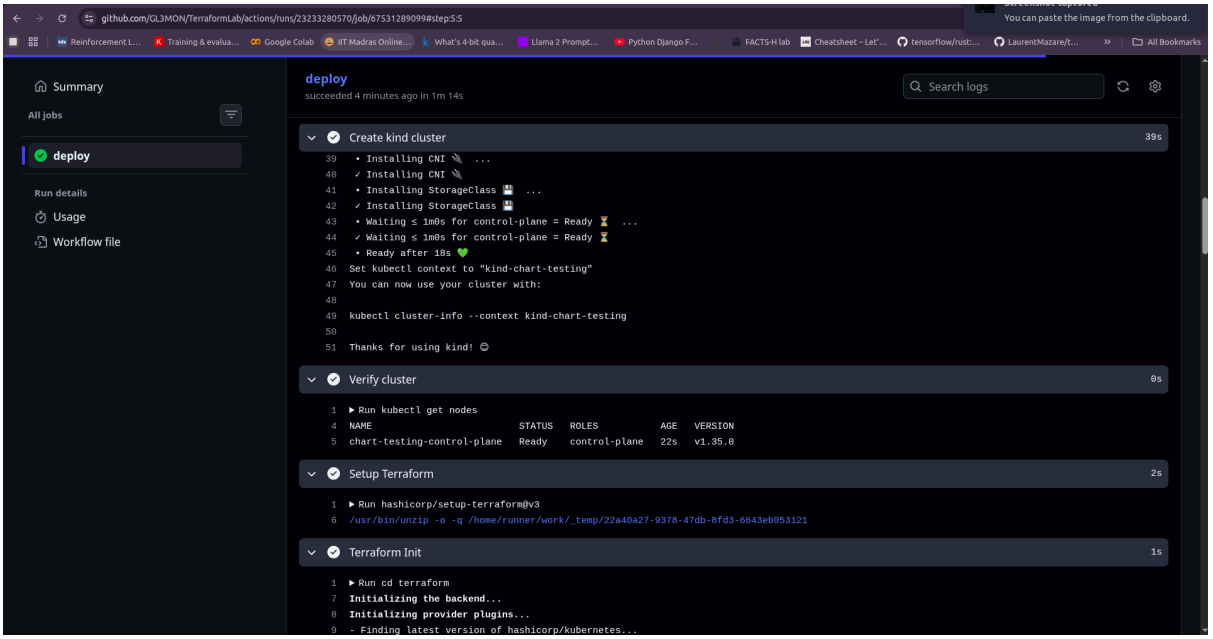
```yaml
    - name: Set up Docker
      uses: docker/setup-buildx-action@v3

    # ⚡ Create Kubernetes cluster (kind)
    - name: Create kind cluster
      uses: helm/kind-action@v1

    # Setup kubectl (auto configured with kind)
    - name: Verify cluster
      run: kubectl get nodes

    # Setup Terraform
    - name: Setup Terraform
      uses: hashicorp/setup-terraform@v3

    # Deploy using Terraform
    - name: Terraform Init
      run: |
        cd terraform
        terraform init

    - name: Terraform Apply
      run: |
        cd terraform
        terraform apply -auto-approve

    # Verify deployment
    - name: Check resources
      run: |
        kubectl get pods -A
```

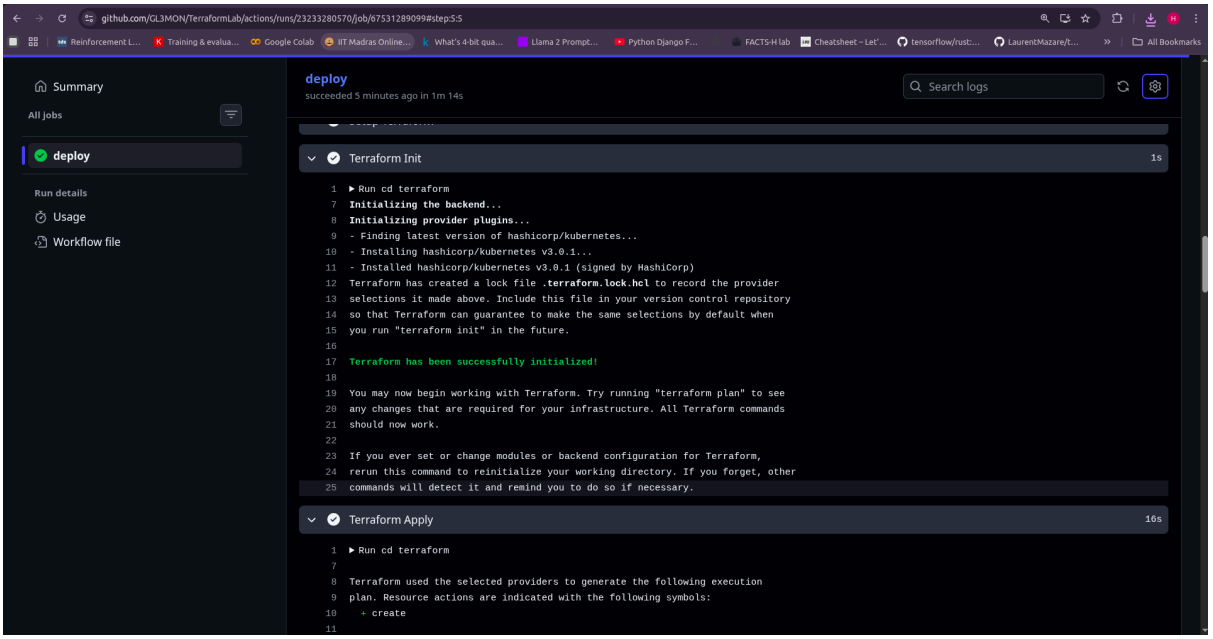

github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5
GL3MON / TerraformLab
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
CI/CD Terraform Deploy
Fixed Deployment #4
Re-run all jobs
Summary
All jobs
deploy
Run details
Usage
Workflow file
Annotations
1 warning
deploy
succeeded 4 minutes ago in 1m 14s
Search logs
Set up job
1s
1 Current runner version: '2.332.0'
2 ▶ Runner Image Provisioner
9 ▶ Operating System
13 ▶ Runner Image
18 ▶ GITHUB_TOKEN Permissions
22 Secret source: Actions
23 Prepare workflow directory
24 Prepare all required actions
25 Getting action download info
26 Download action repository 'actions/checkout@v3' (SHA:f43a0e5ff2bd294095638e18286ca9a3d1956744)
27 Download action repository 'docker/setup-buildx-action@v3' (SHA:8d2750c68a42422c14e847fe6c8ac0403b4cbd6f)
28 Download action repository 'helm/kind-action@v1' (SHA:ef37e7f390d99f746eb8b610417061a60e82a6cc)
29 Download action repository 'hashicorp/setup-terraform@v3' (SHA:b9cd54a3c349d3f38e8881555d616ced269862dd)
30 Complete job name: deploy

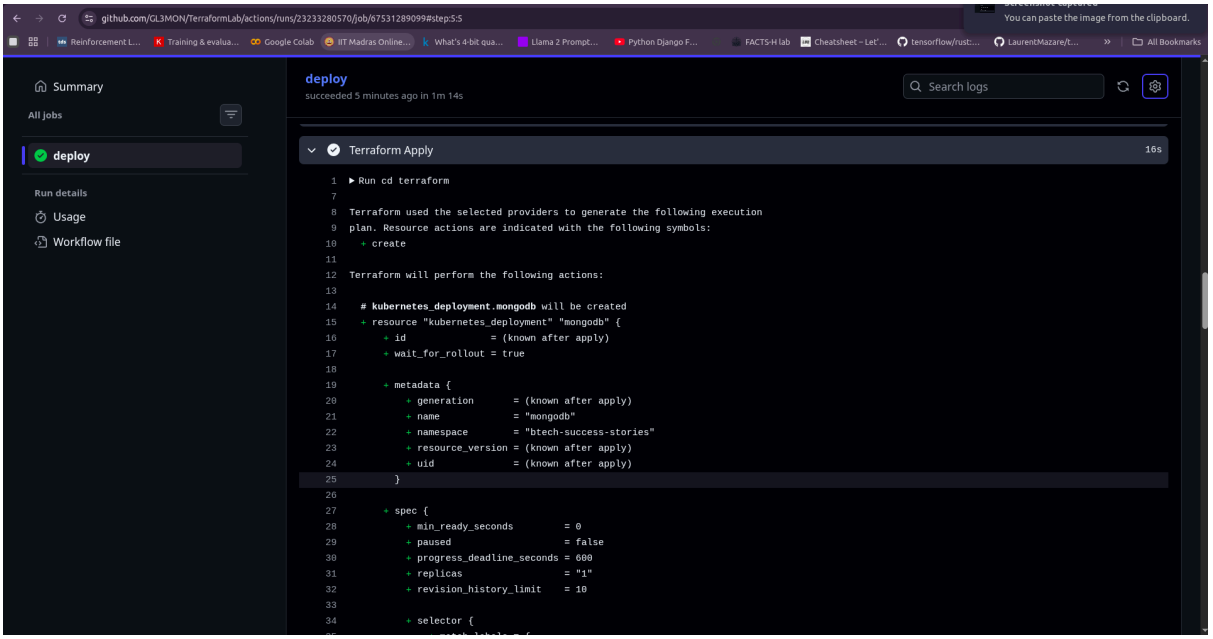

github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5
Summary
All jobs
deploy
Run details
Usage
Workflow file
deploy
succeeded 4 minutes ago in 1m 14s
Search logs
Set up job
1s
1 Current runner version: '2.332.0'
2 ▶ Runner Image Provisioner
9 ▶ Operating System
13 ▶ Runner Image
18 ▶ GITHUB_TOKEN Permissions
22 Secret source: Actions
23 Prepare workflow directory
24 Prepare all required actions
25 Getting action download info
26 Download action repository 'actions/checkout@v3' (SHA:f43a0e5ff2bd294095638e18286ca9a3d1956744)
27 Download action repository 'docker/setup-buildx-action@v3' (SHA:8d2750c68a42422c14e847fe6c8ac0403b4cbd6f)
28 Download action repository 'helm/kind-action@v1' (SHA:ef37e7f390d99f746eb8b610417061a60e82a6cc)
29 Download action repository 'hashicorp/setup-terraform@v3' (SHA:b9cd54a3c349d3f38e8881555d616ced269862dd)
30 Complete job name: deploy
Checkout
1s
1 ▶ Run actions/checkout@v3
14 Syncing repository: GL3MON/TerraformLab
15 ▶ Getting Git version info
19 Temporarily overriding HOME='/home/runner/work/_temp/0674f283-7236-47f1-b0f6-0a363452f2c3' before making global git config changes
20 Adding repository directory to the temporary git global config as a safe directory
21 /usr/bin/git config --global --add safe.directory /home/runner/work/TerraformLab/TerraformLab
22 Deleting the contents of '/home/runner/work/TerraformLab/TerraformLab'
23 ▶ Initializing the repository
40 ▶ Disabling automatic garbage collection
42 ▶ Setting up auth
48 ▶ Fetching the repository
75 ▶ Determining the checkout info

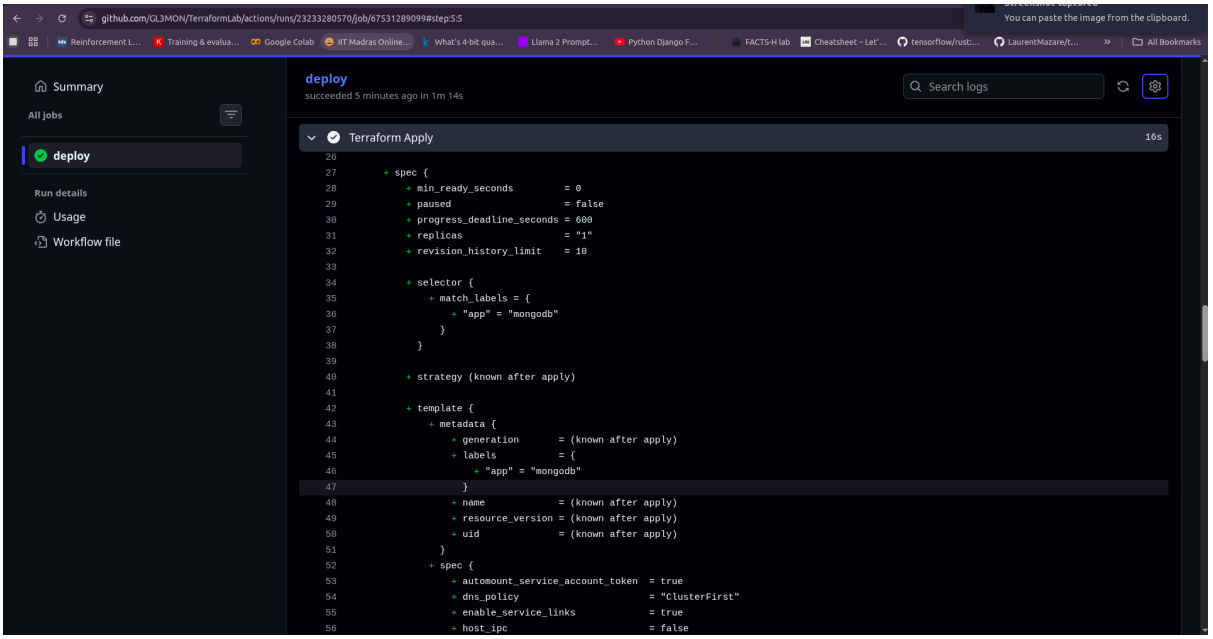

github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5
Summary
All jobs
deploy
Run details
Usage
Workflow file
deploy
succeeded 4 minutes ago in 1m 14s
Search logs
Checkout 1s
76 ▶ Checking out the ref
80 /usr/bin/git log -1 --format='%H'
81 'dd4323629304 6e3679f801d480361fd46c265912'
Set up Docker 8s
1 ▶ Run docker/setup-buildx-action@v3
9 ▶ Docker info
100 ▶ Buildx version
103 ▶ Inspecting default docker context
121 ▶ Creating a new builder instance
124 ▶ Booting builder
171 ▶ Inspect builder
234 ▶ BuildKit version
Create kind cluster 39s
1 ▶ Run helm/kind-action@v1
16 Installing kind...
17 /opt/hostedtoolcache/kind/v0.31.0/amd64/kind/bin ~/work/TerraformLab/TerraformLab
18 kind-linux-amd64: OK
19 ~/work/TerraformLab/TerraformLab
20 Adding kind directory to PATH...
21 Installing kubectl...
22 /opt/hostedtoolcache/kind/v0.31.0/amd64/kubectl/bin ~/work/TerraformLab/TerraformLab
23 kubectl: OK
24 ~/work/TerraformLab/TerraformLab
25 Adding kubectl directory to PATH...
26 kind v0.31.0 go1.25.5 linux/amd64
27 Client Version: v1.35.0


github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5
Summary
All jobs
deploy
Run details
Usage
Workflow file
deploy
succeeded 4 minutes ago in 1m 14s
Search logs
Create kind cluster
39s
Installing kind...
/opt/hostedtoolcache/kind/v0.31.0/amd64/kind/bin ~/work/TerraformLab/TerraformLab
kind-linux-amd64: OK
~/work/TerraformLab/TerraformLab
Adding kind directory to PATH...
Installing kubectl...
/opt/hostedtoolcache/kind/v0.31.0/amd64/kubectl/bin ~/work/TerraformLab/TerraformLab
kubectl: OK
~/work/TerraformLab/TerraformLab
Adding kubectl directory to PATH...
kind v0.31.0 go1.25.5 linux/amd64
Client Version: v1.35.0
Kustomize Version: v5.7.1
Creating kind cluster...
Creating cluster "chart-testing" ...
• Ensuring node image (kindest/node:v1.35.0) ...
✓ Ensuring node image (kindest/node:v1.35.0)
• Preparing nodes ...
✓ Preparing nodes
• Writing configuration ...
✓ Writing configuration
• Starting control-plane ...
✓ Starting control-plane
• Installing CNI ...
✓ Installing CNI
• Installing StorageClass ...
✓ Installing StorageClass
• Waiting ≤ 1m0s for control-plane = Ready ...
✓ Waiting ≤ 1m0s for control-plane = Ready
• Ready after 18s


github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5
You can paste the image from the clipboard.
Summary
All jobs
deploy
Run details
Usage
Workflow file
deploy
succeeded 4 minutes ago in 1m 14s
Search logs
Create kind cluster
39s
• Installing CNI ...
✓ Installing CNI
• Installing StorageClass ...
✓ Installing StorageClass
• Waiting ≤ 1m0s for control-plane = Ready ...
✓ Waiting ≤ 1m0s for control-plane = Ready
• Ready after 18s
Set kubectl context to "kind-chart-testing"
You can now use your cluster with:
kubectl cluster-info --context kind-chart-testing
Thanks for using kind!
Verify cluster
0s
Run kubectl get nodes
NAME STATUS ROLES AGE VERSION
chart-testing-control-plane Ready control-plane 22s v1.35.0
Setup Terraform
2s
Run hashicorp/setup-terraform@v3
/usr/bin/unzip -o -q /home/runner/work/_temp/22a40a27-9378-47db-8fd3-6643eb053121
Terraform Init
1s
Run cd terraform
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/kubernetes...


github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5
Summary
All jobs
deploy
Run details
Usage
Workflow file
deploy
succeeded 5 minutes ago in 1m 14s
Search logs
Terraform Init
1s
Run cd terraform
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/kubernetes...
- Installing hashicorp/kubernetes v3.0.1...
- Installed hashicorp/kubernetes v3.0.1 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
Terraform has been successfully initialized!
You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
Terraform Apply
16s
Run cd terraform
Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
+ create


github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5
You can paste the image from the clipboard.
Reinforcement L...
Training & evalua...
Google Colab
IIT Madras Online...
What's 4-bit qua...
Llama 2 Prompt...
Python Django F...
FACTS-H lab
Cheatsheet – Let'...
tensorflow/rust:...
LaurentMazare/t...
All Bookmarks
Summary
All jobs
deploy
Run details
Usage
Workflow file
deploy
succeeded 5 minutes ago in 1m 14s
Search logs
Terraform Apply
16s
1 ▶ Run cd terraform
7
8 Terraform used the selected providers to generate the following execution
9 plan. Resource actions are indicated with the following symbols:
10 + create
11
12 Terraform will perform the following actions:
13
14 # kubernetes_deployment.mongodb will be created
15 + resource "kubernetes_deployment" "mongodb" {
16 + id = (known after apply)
17 + wait_for_rollout = true
18
19 + metadata {
20 + generation = (known after apply)
21 + name = "mongodb"
22 + namespace = "btech-success-stories"
23 + resource_version = (known after apply)
24 + uid = (known after apply)
25 }
26
27 + spec {
28 + min_ready_seconds = 0
29 + paused = false
30 + progress_deadline_seconds = 600
31 + replicas = "1"
32 + revision_history_limit = 10
33
34 + selector {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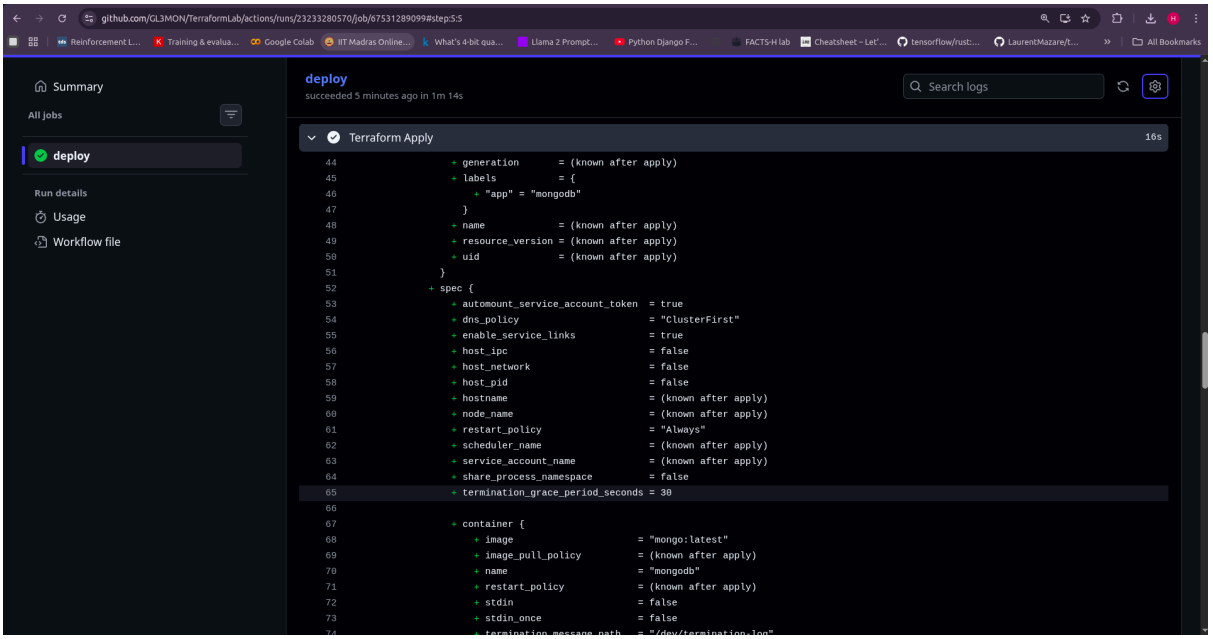
26
27 + spec {
28 + min_ready_seconds = 0
29 + paused = false
30 + progress_deadline_seconds = 600
31 + replicas = "1"
32 + revision_history_limit = 10
33
34 + selector {
35 + match_labels = {
36 + "app" = "mongodb"
37 }
38 }
39
40 + strategy (known after apply)
41
42 + template {
43 + metadata {
44 + generation = (known after apply)
45 + labels = {
46 + "app" = "mongodb"
47 }
48 + name = (known after apply)
49 + resource_version = (known after apply)
50 + uid = (known after apply)
51 }
52 + spec {
53 + automount_service_account_token = true
54 + dns_policy = "ClusterFirst"
55 + enable_service_links = true
56 + host_ipc = false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Summary
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deploy
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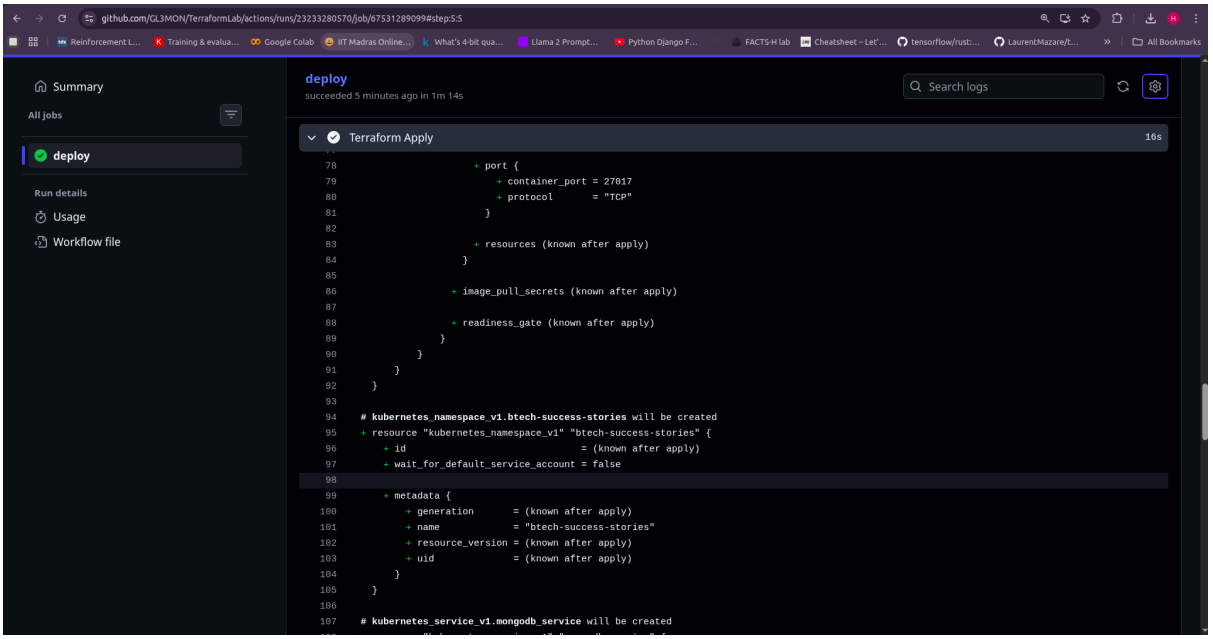
deploy
succeeded 5 minutes ago in 1m 14s
Search logs
Terraform Apply
16s
44 + generation = (known after apply)
45 + labels = {
46 + "app" = "mongodb"
47 }
48 + name = (known after apply)
49 + resource_version = (known after apply)
50 + uid = (known after apply)
51 }
52 + spec {
53 + automount_service_account_token = true
54 + dns_policy = "ClusterFirst"
55 + enable_service_links = true
56 + host_ipc = false
57 + host_network = false
58 + host_pid = false
59 + hostname = (known after apply)
60 + node_name = (known after apply)
61 + restart_policy = "Always"
62 + scheduler_name = (known after apply)
63 + service_account_name = (known after apply)
64 + share_process_namespace = false
65 + termination_grace_period_seconds = 30
66
67 + container {
68 + image = "mongo:latest"
69 + image_pull_policy = (known after apply)
70 + name = "mongodb"
71 + restart_policy = (known after apply)
72 + stdin = false
73 + stdin_once = false


github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5
Summary
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succeeded 5 minutes ago in 1m 14s
Search logs
Terraform Apply
16s
78 + port {
79 + container_port = 27017
80 + protocol = "TCP"
81 }
82
83 + resources (known after apply)
84 }
85
86 + image_pull_secrets (known after apply)
87
88 + readiness_gate (known after apply)
89 }
90 }
91 }
92 }
93
94 # kubernetes_namespace_v1.btech-success-stories will be created
95 + resource "kubernetes_namespace_v1" "btech-success-stories" {
96 + id = (known after apply)
97 + wait_for_default_service_account = false
98
99 + metadata {
100 + generation = (known after apply)
101 + name = "btech-success-stories"
102 + resource_version = (known after apply)
103 + uid = (known after apply)
104 }
105 }
106
107 # kubernetes_service_v1.mongodb_service will be created

github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5

Summary

All jobs

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succeeded 5 minutes ago in 1m 14s

Search logs

Terraform Apply 16s

```
106
107   # kubernetes_service_v1.mongodb_service will be created
108   + resource "kubernetes_service_v1" "mongodb_service" {
109       + id                     = (known after apply)
110       + status                 = (known after apply)
111       + wait_for_load_balancer = true
112
113       + metadata {
114           + generation       = (known after apply)
115           + name             = "mongodb-service"
116           + namespace        = "btech-success-stories"
117           + resource_version = (known after apply)
118           + uid              = (known after apply)
119         }
120
121       + spec {
122           + allocate_load_balancer_node_ports = true
123           + cluster_ip                        = (known after apply)
124           + cluster_ips                       = (known after apply)
125           + external_traffic_policy           = (known after apply)
126           + health_check_node_port            = (known after apply)
127           + internal_traffic_policy           = (known after apply)
128           + ip_families                       = (known after apply)
129           + ip_family_policy                  = (known after apply)
130           + publish_not_ready_addresses       = false
131           + selector                          = {
132               + "app" = "mongodb"
133             }
134           + session_affinity                  = "None"
135           + type                              = "ClusterIP"
136
```

github.com/GL3MON/TerraformLab/actions/runs/23233280570/job/67531289099#step:5:5

Summary

All jobs

deploy

Run details

Usage

Workflow file

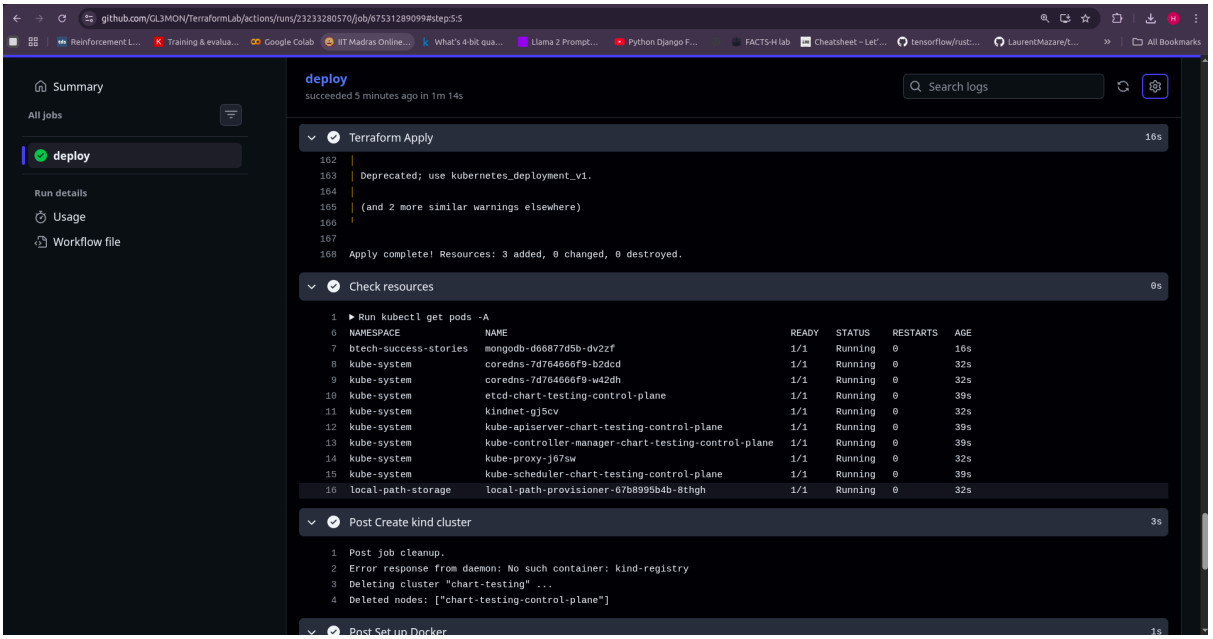
deploy

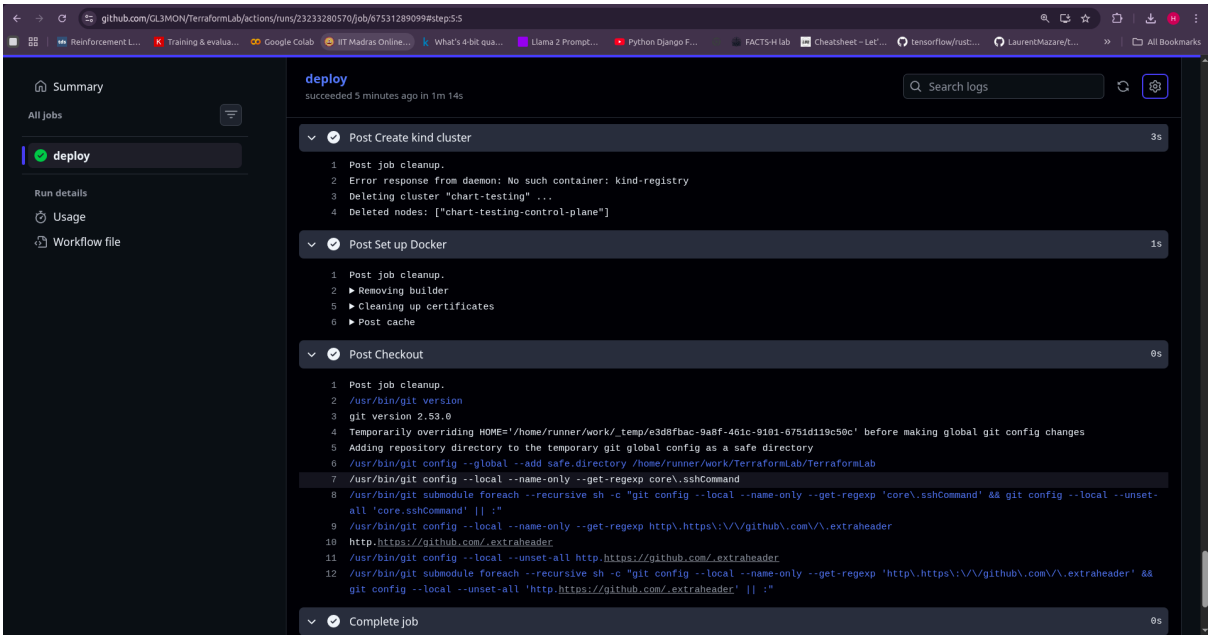
succeeded 5 minutes ago in 1m 14s

Search logs

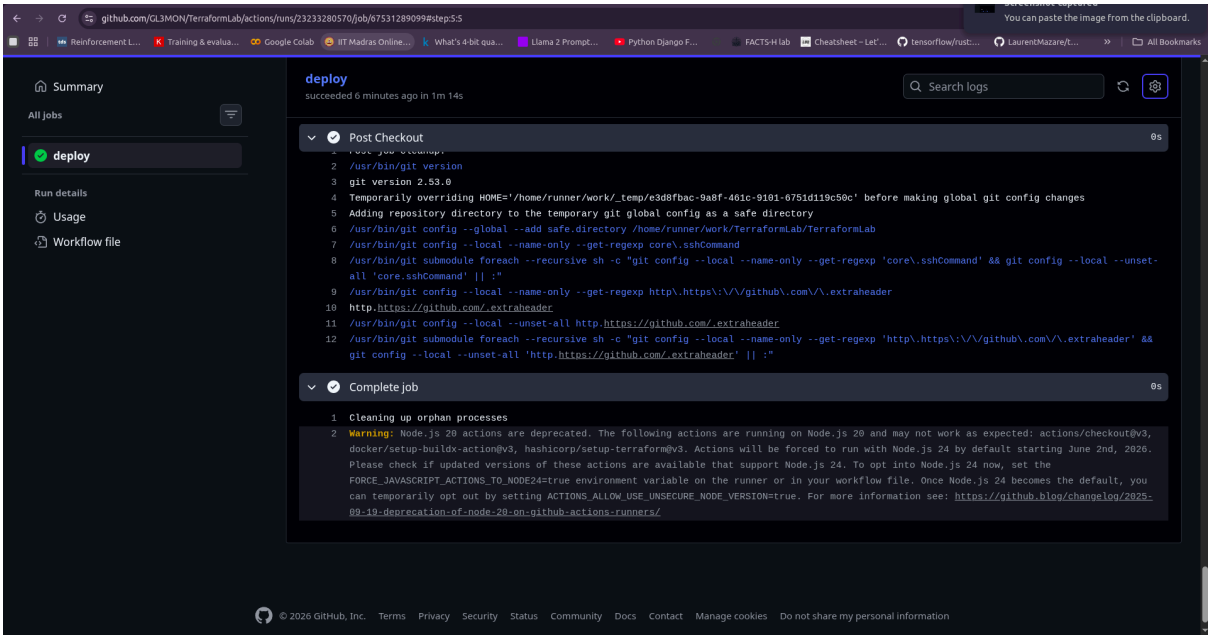
Terraform Apply 16s

```
137           + port {
138               + node_port   = (known after apply)
139               + port        = 27017
140               + protocol    = "TCP"
141               + target_port = "27017"
142             }
143
144           + session_affinity_config (known after apply)
145         }
146     }
147
148 Plan: 3 to add, 0 to change, 0 to destroy.
149 kubernetes_namespace_v1.btech-success-stories: Creating...
150 kubernetes_namespace_v1.btech-success-stories: Creation complete after 0s [id=btech-success-stories]
151 kubernetes_service_v1.mongodb_service: Creating...
152 kubernetes_deployment.mongodb: Creating...
153 kubernetes_service_v1.mongodb_service: Creation complete after 0s [id=btech-success-stories/mongodb-service]
154 kubernetes_deployment.mongodb: Still creating... [00m10s elapsed]
155 kubernetes_deployment.mongodb: Creation complete after 15s [id=btech-success-stories/mongodb]
156 |
157 | Warning: Deprecated Resource
158 |
159 |   with kubernetes_deployment.mongodb,
160 |   on main.tf line 14, in resource "kubernetes_deployment" "mongodb":
161 |   14: resource "kubernetes_deployment" "mongodb" {
162 |
163 | Deprecated; use kubernetes_deployment_v1.
164 |
165 | (and 2 more similar warnings elsewhere)
166 |
167
```





The CI/CD pipeline via github actions successfully builds the image and creates a kubernetes cluster using kind via terraform. So, every change in the code will update the cluster.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deploy
succeeded 6 minutes ago in 1m 14s
Search logs
Post Checkout
0s
/usr/bin/git version
git version 2.53.0
Temporarily overriding HOME='/home/runner/work/_temp/e3d8fbac-9a8f-461c-9101-6751d119c50c' before making global git config changes
Adding repository directory to the temporary git global config as a safe directory
/usr/bin/git config --global --add safe.directory /home/runner/work/TerraformLab/TerraformLab
/usr/bin/git config --local --name-only --get-regexp core\.sshCommand
/usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'core\.sshCommand' && git config --local --unset-all 'core.sshCommand' || :"
/usr/bin/git config --local --name-only --get-regexp http\.https\:\/\/github\.com\/\.extraheader
http.https://github.com/.extraheader
/usr/bin/git config --local --unset-all http.https://github.com/.extraheader
/usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'http\.https\:\/\/github\.com\/\.extraheader' && git config --local --unset-all 'http.https://github.com/.extraheader' || :"
Complete job
0s
Cleaning up orphan processes
Warning: Node.js 20 actions are deprecated. The following actions are running on Node.js 20 and may not work as expected: actions/checkout@v3, docker/setup-buildx-action@v3, hashicorp/setup-terraform@v3. Actions will be forced to run with Node.js 24 by default starting June 2nd, 2026. Please check if updated versions of these actions are available that support Node.js 24. To opt into Node.js 24 now, set the FORCE_JAVASCRIPT_ACTIONS_TO_NODE24=true environment variable on the runner or in your workflow file. Once Node.js 24 becomes the default, you can temporarily opt out by setting ACTIONS_ALLOW_USE_UNSECURE_NODE_VERSION=true. For more information see: https://github.blog/changelog/2025-09-19-deprecation-of-node-20-on-github-actions-runners/
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